

Operating Systems — Lab

Assignment 1: Outputs

1. uptime.c

```
aadeshkp@aadeshkp: ~/Misc/xv6-labs-2025
riscv64-unknown-elf-objdump -S user/_uptime > user/uptime.asm
riscv64-unknown-elf-objdump -t user/_uptime | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$/d' > user/uptime.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_uptime user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1137 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
uptime
166 ticks
```

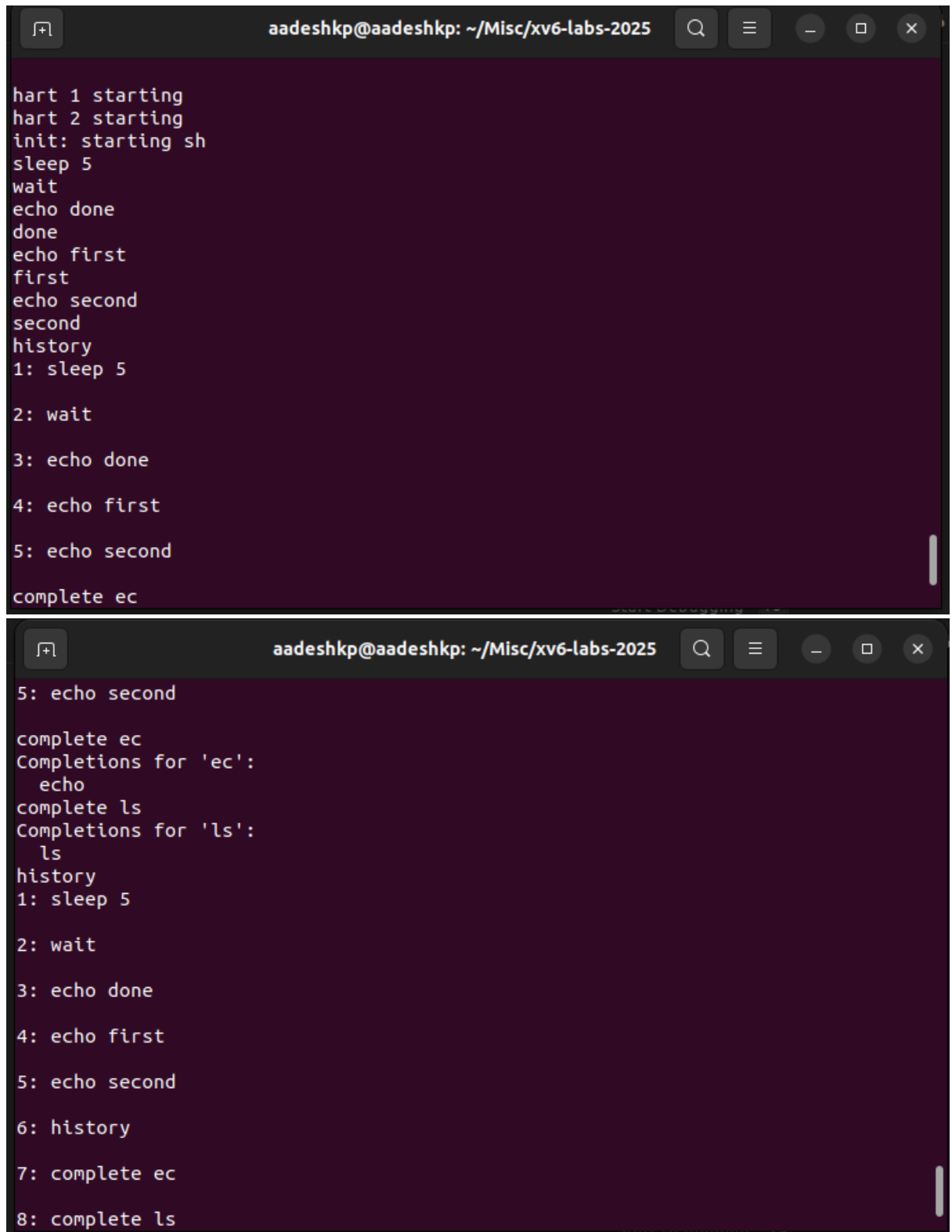
2. find.c (regex support)

```
aadeshkp@aadeshkp: ~/Misc/xv6-labs-2025
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
uptime
166 ticks
find / README
//README
find / sh
//findtest.sh
//sh
find / txt
//sixfive.txt
find / c
//cat
//echo
//wc
```

3. sh.c



```
aadeshkp@aadeshkp: ~/Misc/xv6-labs-2025

hart 1 starting
hart 2 starting
init: starting sh
sleep 5
wait
echo done
done
echo first
first
echo second
second
history
1: sleep 5

2: wait

3: echo done

4: echo first

5: echo second

complete ec

5: echo second

complete ec
Completions for 'ec':
  echo
complete ls
Completions for 'ls':
  ls
history
1: sleep 5

2: wait

3: echo done

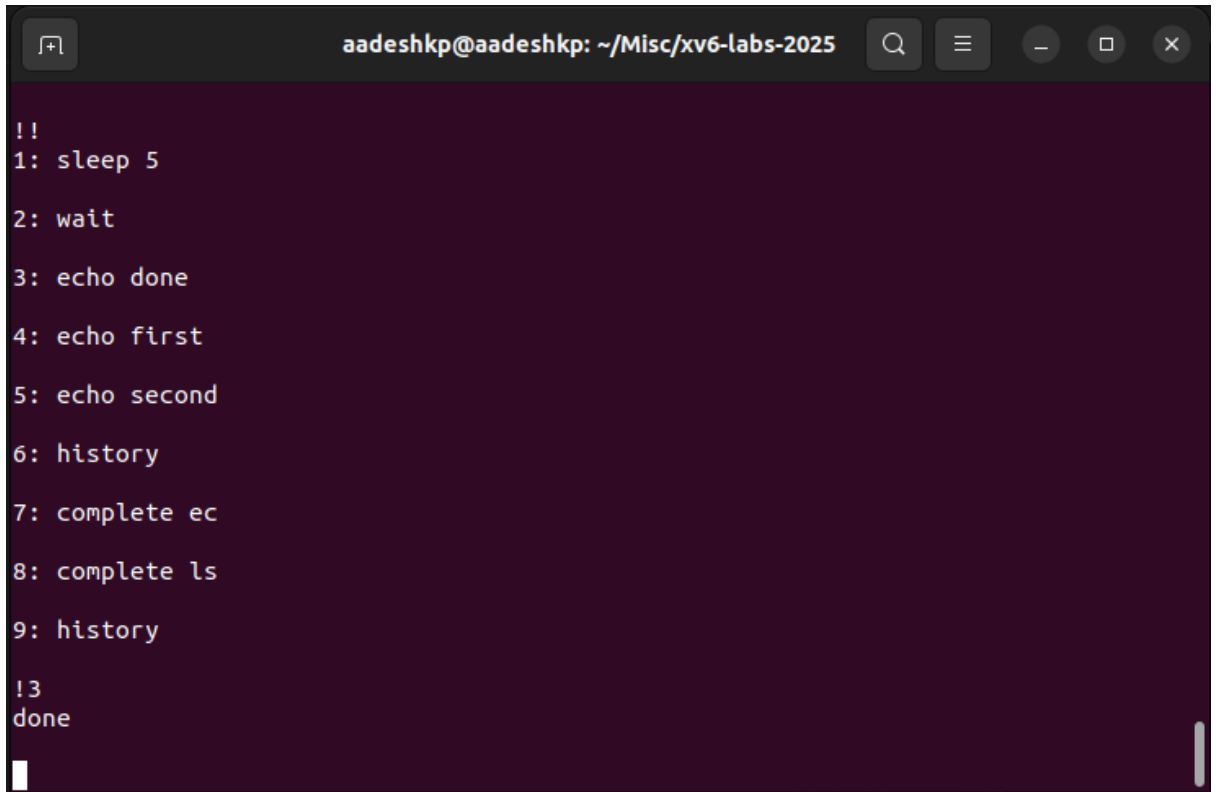
4: echo first

5: echo second

6: history

7: complete ec

8: complete ls
```



The screenshot shows a terminal window with a dark background and light-colored text. The window title bar at the top reads "aadeshkp@aadeshkp: ~/Misc/xv6-labs-2025". The terminal content shows a series of commands being executed in a shell script, numbered 1 through 9. The commands are: 1: sleep 5, 2: wait, 3: echo done, 4: echo first, 5: echo second, 6: history, 7: complete ec, 8: complete ls, and 9: history. The output of the script is displayed at the bottom, showing "13" and "done".

```
aadeshkp@aadeshkp: ~/Misc/xv6-labs-2025

!!
1: sleep 5
2: wait
3: echo done
4: echo first
5: echo second
6: history
7: complete ec
8: complete ls
9: history

13
done
```

Changelog in sh.c:

- Changelog
- No \$ for non-interactive mode.
- Implemented WAIT
- Tab completion via COMPLETE
- Command history via HISTORY, !!, !n